MIL-C-45686(Ord)
7 February 1961
SUPERSEDING
FA(FC)-PD-704A
10 February 1960

MILITARY SPECIFICATION

CASES, FIRE CONTROL, ELECTRONIC CALIBRATION STANDARDS: REUSABLE, METAL

1. SCOPE

1.1 Scope. - This specification covers multipurpose, reusable, rectangular metal cases for shipment of electronic test equipment to and from calibration points, and for shipment and storage of electronic test equipment requiring water vaporproof protection. (See 6.1)

1.2 Classification. - Cases shall be of the following types:

Type I	For 19 Inch Rack Mounted Units (Not exceeding 17 inches high, by 14 inches deep)
Type II	For 19 Inch Rack Mounted Units (Not exceeding 10 inches high, by 11 inches deep)
Type III	For Cabinet Mounted Units (Not exceeding 13 inches wide, by 9 inches high, by 9 inches deep)
Type IV	For Fully Encased Units (Not exceeding 16 inches wide, by 5 inches high, by 10 inches deep)

FSC 1290

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2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-A-5092	Adhesive, Rubber (Synthetic and Reclaimed Rubber Base)
MIL-C-7769	Cushioning Material, Bound
MIL-W-8604	Welding of Aluminum Alloys: Process For
MIL-I-45208	Inspection Requirements; General Specification For

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage

DRAWINGS

Ordnance Corps

F8293100	Box, Metal, Shipping

(Copies of specifications, standards, and drawings required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.

3. REQUIREMENTS

3.1 Production sample. - Prior to production, one case, representative of production, shall be made available to the Government to determine if the case meets all requirements of this specification.

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3.2 Material. - Material shall be as specified herein and as shown on applicable drawings. Material shall be free from defects that might affect serviceability of the finished containers. Unless otherwise specified, the contractor shall provide certification of compliance of the material specifications.

3.3 Fabrication . - Each case shall be fabricated in accordance with Drawing F8293100 and drawings pertaining thereto.

3.4 Cushioning. - Homogeneously molded rubberized curled hair inserts shall be provided within each case. The inserts shall be fabricated from material conforming to Specification MIL-C-7769 and shall conform to the density and dimensional requirements of the applicable cushion drawing. The binding material shall be neoprene or approved substantially equal.

3.4.1 Securing molded inserts. - The surface of the molded cushion supports in contact with the interior of the cases shall be firmly secured in place with adhesive conforming to Type II of Specification MIL-A-5092.

3.5 Welding. - Welds shall be reasonably smooth and free of craters. They shall exhibit characteristics of fusion, penetration and soundness of good welding practice and shall meet the requirements of Specification MIL-W-8604. Parts to be welded shall be clean, free from rust, scale, corrosion, oil and other foreign substances before welding operations are performed.

3.6 Breather valves. - Breather valves shall be set so that they have positive action to maintain 2.5 psi pressure and 1.5 psi vacuum in the case.

3.7 Closure. - Closure shall be as specified on the applicable drawing. The gasket shall be embedded in the bottom flange in such a manner as to provide proper compression required for sealing. The gasket shall be properly secured and shall not permit leakage when the case is subjected to the tests specified in Section 4. There shall be no excessive adhesive remaining in the closure joint. Fasteners provided on cases shall be capable of repeated opening and closure with normal hand operation.

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3.8 Leakage.- Prior to the application of the required finishing process and prior to shipment, each case shall be subjected to the leak test specified in Section 4 or approved alternate test and shall show no evidence of leakage. The case should be capable of withstanding 2.0 plus or minus .25 psi for a duration of at least 5 minutes.

3.9 Workmanship.- Workmanship shall be of the quality necessary to produce containers free from defects that might affect their strength, serviceability and durability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Contractor's quality assurance system.- The contractor shall provide and maintain an adequate quality assurance system acceptable to the Government covering the supplies under the contract. A current written description of the system shall be submitted to the contracting officer prior to initiation of production. The written description will be considered acceptable when it provides:

(a) The quality assurance required by the applicable drawings and specifications listed in the contract or order.

(b) Contractor's quality standards of acceptability.

(c) Procedure for continually evaluating product quality.

(d) Procedure for continually reporting evidence of actual product quality.

(e) Measures for preventing any significant departure of product quality from standards.

4.1.1 The contractor shall notify the Government of,and obtain approval for, any change to the written procedure that might affect the degree of assurance required by this specification and other applicable documents referenced herein.

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4.2 Government verification. - All quality assurance operations performed bythe contractor will be subject to Government verification at unscheduled intervals. Verification shall consist of: (a) Surveillance of the operations to determine whether the practices, methods, and procedures of the written plan are being properly applied, and (b) Government product inspection to measure quality of product offered for acceptance. Deviations from the prescribed or agreed-upon procedures, or instances of poor practices which might have an affect upon the quality of the product will be immediately called to the attention of the contractor. Failure of the contractor to promptly correct deficiencies discovered shall be cause for suspension of acceptance until corrections have been made or until conformance of product to prescribed criteria has been demonstrated. To avoid interference with operations, the contractor shall designate a responsible official or officials to whom the Government inspector will report such instances.

4.3 Contractor's inspection system. - Contractor's inspection and Government verification of contractor's inspection shall be in accordance with Specification MIL-I-45208.

4.4 Production sample. - Unless otherwise specified in the contract, one case shall be selected by the Government inspector as soon as practicable after the start of production, and be inspected by the contractor under the surveillance of the Government. This inspection shall be conducted at the contractor's facility, or Government approved laboratory, using test equipment authorized for use on the contract. The sample shall be inspected for all requirements of this specification by utilizing examination and testing methods prescribed herein.

4.4.1 Production sample failure. - Should the production sample fail to pass the prescribed tests, the contractor shall cease those operations which are causing the rejection and shall make the necessary corrections. When corrections in the process have been made, a new production sample shall be examined and tested by the contractor under the surveillance of the Government. Acceptance of production lots will not commence until an acceptable production sample is submitted and corrective action has been taken to eliminate deficiencies.

4.5 Lot formation. - Inspection lot sizes and lot formation shall be in accordance with Standard MIL-STD-105.

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4.5.1 Defective units and lots shall be processed as specified in Standard MIL-STD-105.

4.6 Material. - Control for requirements of material shall be in accordance with Specification MIL-I-45208.

4.7 Tests.

4.7.1 Classification of defects. - The classification of defects in Table I shall constitute the minimum inspection requirements to be performed prior to acceptance or rejection by lots. The Government reserves the right to inspect for any or all applicable requirements and to reject individual non-conforming items.

4.7.1.1 Disposition of rejected lots. - Rejected lots shall be screened for all defective characteristics. After corrections have been made and the Government inspector informed of the corrective action taken, the lots shall be re-submitted for acceptance as outlined in Standard MIL-STD-105.

TABLE I - CLASSIFICATION OF DEFECTS

Use Inspection Level II, and Sampling Plan Table IV-A of Standard MIL-STD-105

CRITICAL: None Defined

MAJOR: AQL 1.5%	Req't Par.	Test Method
101. Completeness of Assembly	3.3	Visual
102. Proper length	Per Dwg.	Standard Measuring Equipment (SME)
103. Proper width	Per Dwg.	SME
104. Proper height	Per Dwg.	SME
105. Proper operation of clamps	3.7	Tactile
106. Proper welding	3.5	MIL-W-8604
107. Proper closing of case	3.7	Tactile
108. Proper setting of valves	3.8	SME
109. Workmanship	3.9	Visual
110. Proper marking	Per Dwg.	Visual
111. Proper finish	Per Dwg.	Visual

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TABLE I - CLASSIFICATION OF DEFECTS (Cont'd)

MINOR: AQL 4.0%	Req't Par.	Test Method
201. Proper shape and size of inserts	8293107 8293105 8293103 8293106 8293101	SME
202. Proper securing of inserts	3.4.1	Tactile

4.8 Inspection equipment. - Supply calibration, maintenance and use of inspection equipment shall be in accordance with Specification MIL-I-45208.

4.8.1 Accuracy requirements. - Inspection equipment which incorporates features for measuring values or sizes established by product requirements shall be capable of measuring these values within an accuracy of ten percent of the specified tolerance, except as otherwise specified. In cases where the ten percent accuracy imposes impractical or very difficult design and fabrication problems, or results in a test equipment cost factor which is exorbitant in relation to the cost and application of the product to be inspected, the Government shall be expeditiously advised in order that necessary appropriate action can be taken.

4.9 Fabrication. - Proper fabrication in accordance with the pertinent drawings can be determined by utilizing standard measuring equipment and visual inspection techniques.

4.10 Cushioning. - By means of tactile inspection techniques, proper binding of the cushioning shall be determined.

4.11 Welding. - The inspection techniques and procedures outlined in Specification MIL-W-8604 shall be used to determine acceptability of all welding.

4.12 Breather valve settings. - The pressure differential setting of the valve may be checked as outlined herein using the following components.

4.12.1 To a filtered air line, connect a pressure regulator, needle valve, pressure gage and hose with a rubber tipped nozzle in that order.

4.12.2 Set the regulator to the "OFF" position and insert the rubber tipped nozzle into the intake valve opening (minimum air flow valve) and open the needle valve.

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4.12.3 Slowly adjust the regulator by the means provided until the minimum air flow valve opens. While making this adjustment, observe the pressure gage and note the pressure reading prior to the valve opening. Valve opening will be indicated by a drop in pressure reading. The pressure reading prior to the drop in pressure will be the pressure setting of the minimum air flow valve. This reading must coincide with the value specified in 3.6.

4.12.4 Having determined that the minimum air valve opens within the proper setting, continue to increase the air pressure by means of the regulator until the pressure reading indicates the value setting for the exhaust airflow. This setting will be determined by observing the pressure reading on the gage as the regulator is slowly adjusted to increase the air pressure. When the pressure reading drops, the maximum airflow valve is open. This pressure reading must coincide with the setting specified in 3.6.

4.13 Container leakage. - Container leakage will be determined by using the set-up and procedure described above. However, the pressure applied for this test shall be set to coincide with the pressure specified in 3.8.

4.13.1 When the pressure specified is applied to the minimum airflow valve, the needle valve shall be closed and the pressure gage shall be observed. The duration of the applied pressure shall be as specified in 3.8. A decline in pressure indicates a leak within the case.

4.14 Closure. - Proper closure (sealing) specified in 3.7, will be assured when the case has been subjected to and successfully passed the leakage test specified in 4.7.6.

4.14.1 By means of visual inspection, determine that the gasket is properly sealed and cleaned as specified in 3.7.

4.14.2 By means of tactile inspection techniques, determine the proper action of the fasteners; repeat this test for the number of times stipulated in 3.7.

4.15 Workmanship shall be in accordance with the establizhed practices for manufacturers of similar or equivalent type cases.

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing and marking. - Each case shall be assembled and fastened in its normal closed position. Marking shall be in accordance with Standard MIL-STD-129.

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6. NOTES

6.1 Intended use. - Containers manufactured in accordance with this specification are intended for, but not limited to, the shipment and storage of the following electronic assemblies:

TYPE I

Nomenclature	Mfgr's No.	Container Part No.
Spectrum Analyzer (Amp and Power Supply)	FXR 2800A	8293111
Spectrum Analyzer (Amp and Power Supply)	PRD 850	8293111
Electronic Counter	HP 522B	8293111
Pulse Generator	HP 212A	8293111
Test Oscillator	HP 650A	8293111
Oscilloscope	LA 239C	8293111
Signal Audio Generator	HP 205AG	8293111
Universal Counter & Timer	Berkley 5500	8293111

TYPE II

Nomenclature	Mfgr's No.	Container Part No.
Spectrum Analyzer (S Band R.F. Head)	FXR S900B	8293112
Spectrum Analyzer (S Band R.F. Head)	PRD 851	8293112
Audio Oscillator	HP 200C	8293112
AC Voltmeter	Ballantype 300	8293112
Radligner	Kay Elec	8293112
Wide Band Amplifier	HP 460B	8293112
Resistance Limit Bridge	GR 1652A	8293112
Attenuator	KS 14243	8293112

TYPE III

Nomenclature	Mfgr's No.	Container Part No.
AC Voltmeter	Ballantyne 300 U/18	8293113
AC Voltmeter	Ballantyne 302	8293113
Voltmeter	Weston 301	8293113
Voltmeter	RCA WV97A	8293113
Multimeter	TS 352AU	8293113
Multimeter	TS 505 A/U	8293113
Megacycle Meter	Meas. Corp. Mod. 59	8293113
Voltmeter AC-DC	HP 410B	8293113

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TYPE IV

Nomenclature	Mfgr's No.	Container Part No.
Oscillograph Test Set Null Voltage	Dumont 301 Industrial Instrument or Leeds & Northrop	8293114 8293114

6.2 Ordering data. - Procurement documents should specify the following:

(a) Title, number and date of this specification
(b) Drawing number and applicable container part number of case.

6.3 Definitions.

6.3.1 Standard measuring equipment (SME). - Standard measuring equipment includes common hand type measuring devices which are usually stocked by commercial supply houses for ready supply (shelf items), and which are normally used by an inspector to perform dimensional inspection of items under procurement. This category also includes commercial testing equipment.

6.3.2 Tactile. - Skillful manipulation or sensing.

NOTICE: When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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